

Derivatives of Vector Functions

In Calc I, the derivative measured **rate of change** of a scalar function.

For a vector function, the same limit definition applies — but the output is now a **vector**.

Derivative of a Vector Function

$$\frac{d\vec{r}}{dt} = \vec{r}'(t) = \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h}$$

If $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$, then:

$$\vec{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$$

Just **differentiate each component** separately.

Physical interpretation: If t is time and $\vec{r}(t)$ is position, then $\vec{r}'(t)$ is the **velocity vector** — it tells you how fast and in which direction you're moving.

Derivative = velocity. The computation is simple: differentiate component by component, just like Calc I.

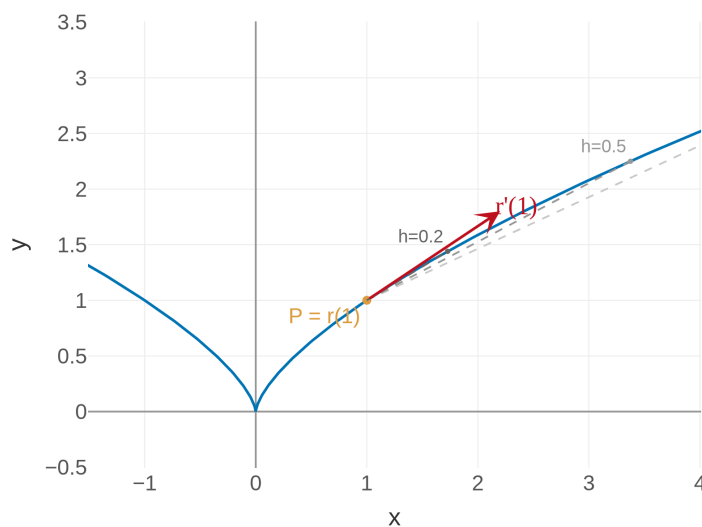
What Does $\vec{r}'(t)$ Mean Geometrically?

In Calc I, the derivative was the slope of the **tangent line**. Here:

The difference $\vec{r}(t+h) - \vec{r}(t)$ points from the current position to a nearby position on the curve.

Dividing by h gives a **secant vector** along the curve.

As $h \rightarrow 0$, the secant vector approaches the **tangent vector** $\vec{r}'(t)$.



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So $\vec{r}'(t)$ is a vector that:

- **Points in the direction** the curve is heading at time t
- Has **magnitude** equal to the speed: $|\vec{r}'(t)| = \text{speed}$

$\vec{r}'(t)$ is the **velocity vector** — direction of travel + speed, all in one package.

The Unit Tangent Vector

Sometimes we only care about the **direction** of travel, not the speed. We normalize:

Unit Tangent Vector

$$\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$$

This is a unit vector ($|\vec{T}| = 1$) that points in the direction of motion.

Think of it this way:

- $\vec{r}'(t)$ = velocity (direction **and** speed)
- $|\vec{r}'(t)|$ = speed (magnitude only)
- $\vec{T}(t)$ = direction only (no speed information)

Dividing by the magnitude strips away speed and isolates pure **direction**.

Example 1: Find $\vec{r}'(t)$ and $\vec{T}(t)$ for $\vec{r}(t) = \langle t^3, t^2 \rangle$ at $t = 1$

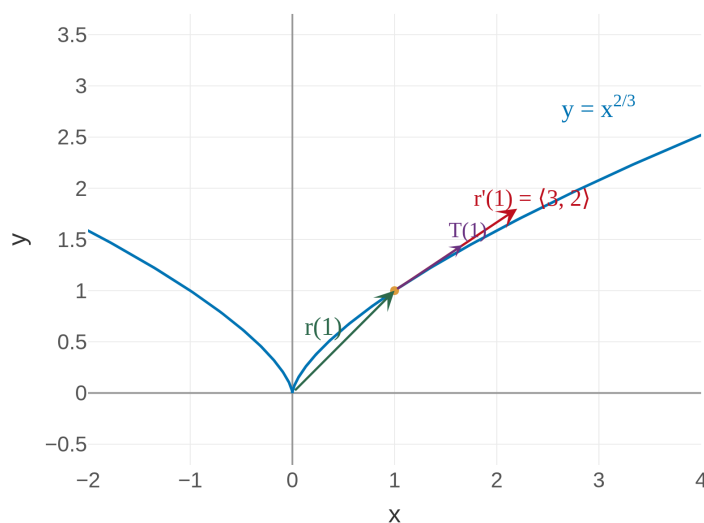
This is the same curve from Section 13.1 — now we're finding its velocity and direction at $t = 1$.

Step 1 — Differentiate component by component:

$$\vec{r}'(t) = \langle 3t^2, 2t \rangle \Rightarrow \vec{r}'(1) = \langle 3, 2 \rangle$$

Step 2 — Find the speed: $|\vec{r}'(1)| = \sqrt{3^2 + 2^2} = \sqrt{13}$

Step 3 — Normalize: $\vec{T}(1) = \frac{\langle 3, 2 \rangle}{\sqrt{13}} = \left\langle \frac{3}{\sqrt{13}}, \frac{2}{\sqrt{13}} \right\rangle$



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At $(1, 1)$, the particle is moving in the direction $\langle 3, 2 \rangle$ with speed $\sqrt{13} \approx 3.6$.

The Velocity Vector as a Prediction

In Calc I, the tangent line was the best linear approximation to $f(x)$ near a point.

Here, the velocity vector predicts where you'll be after a short time step.

Starting at $\vec{r}(1) = \langle 1, 1 \rangle$ with velocity $\vec{r}'(1) = \langle 3, 2 \rangle$:

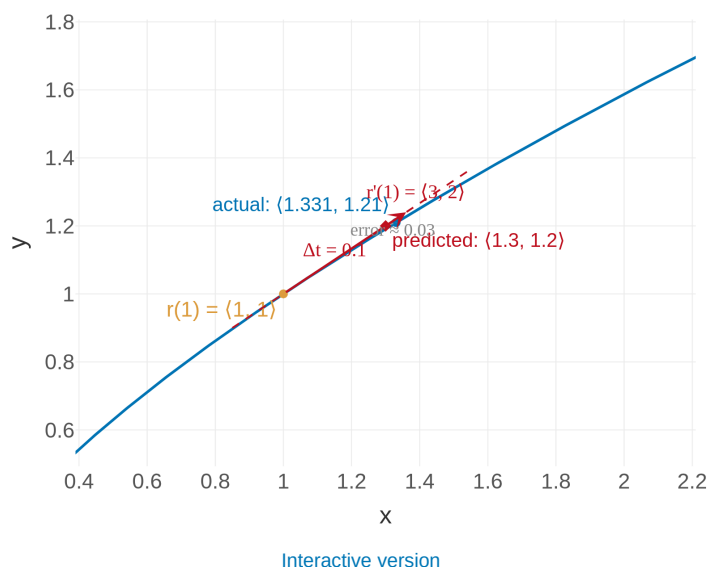
Actual position at $t = 1.1$ — follow the curve:

$$\vec{r}(1.1) = \langle 1.1^3, 1.1^2 \rangle = \langle 1.331, 1.21 \rangle$$

Predicted position — follow the velocity for $\Delta t = 0.1$:

$$\vec{r}(1) + 0.1 \vec{r}'(1) = \langle 1, 1 \rangle + 0.1 \langle 3, 2 \rangle = \langle 1.3, 1.2 \rangle$$

$\langle 1.331, 1.21 \rangle$ vs. $\langle 1.3, 1.2 \rangle$ — almost identical! The velocity vector is the **linear approximation** of motion along the curve.



Your Turn: Find $\vec{r}'(t)$ and $\vec{T}(t)$ for $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$ at $t = 0$

This is the helix from Section 13.1. Differentiate, evaluate, normalize.

Try it before looking at the solution.

Differentiate: $\vec{r}'(t) = \langle -\sin t, \cos t, 1 \rangle$

Evaluate: $\vec{r}'(0) = \langle 0, 1, 1 \rangle$

Speed: $|\vec{r}'(0)| = \sqrt{0 + 1 + 1} = \sqrt{2}$

Normalize: $\vec{T}(0) = \frac{1}{\sqrt{2}} \langle 0, 1, 1 \rangle$

Bonus: $|\vec{r}'(t)| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}$ for **all** t — the helix has constant speed!

Theorem: Constant Distance $\Rightarrow$ Orthogonality

Question: If a particle moves on a sphere (constant distance from the origin), what can we say about its velocity? Think about it: a ball swinging on a string always moves **sideways**, never toward or away from the center.

Orthogonality Theorem

If $|\vec{r}(t)| = c$ (constant), then $\vec{r}(t) \cdot \vec{r}'(t) = 0$.

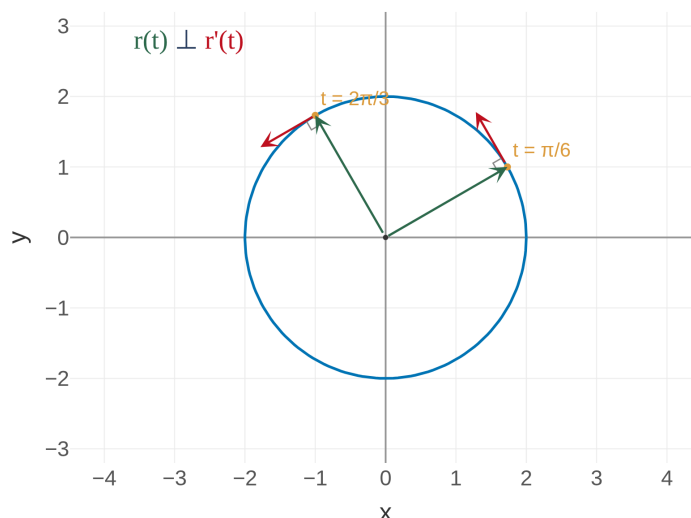
In other words: **position $\perp$ velocity**.

Proof: Start with $|\vec{r}(t)|^2 = c^2$, i.e., $\vec{r}(t) \cdot \vec{r}(t) = c^2$.

Differentiate both sides using the product rule for dot products:

$$\vec{r}'(t) \cdot \vec{r}(t) + \vec{r}(t) \cdot \vec{r}'(t) = 0$$

$$2\vec{r}(t) \cdot \vec{r}'(t) = 0 \Rightarrow \vec{r}(t) \cdot \vec{r}'(t) = 0 \quad \square$$



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On a sphere or circle, velocity is always **perpendicular to the radius**. This is why orbits are tangent to their paths.

Integrals of Vector Functions

Motivation: If $\vec{r}'(t)$ is velocity, what is $\int_a^b \vec{r}'(t) dt$?

Just like in Calc I: integrating velocity gives **displacement** (change in position).

The rule is the same as for derivatives — **component by component**.

Definite Integral of a Vector Function

$$\int_a^b \vec{r}(t) dt = \left\langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \right\rangle$$

For **antiderivatives**, don't forget the constant is now a **vector**:

$$\int \vec{r}(t) dt = \left\langle \int f(t) dt, \int g(t) dt, \int h(t) dt \right\rangle + \vec{C}$$

No new techniques — just Calc I integration applied to each component. The only wrinkle: $\vec{C}$ is a constant **vector**, not a number.

Example 2: Evaluate $\int_1^4 \left(2t^{3/2} \mathbf{i} + \frac{1}{1+t^2} \mathbf{j} + (t+1)\sqrt{t} \mathbf{k} \right) dt$

Integrate each component separately. Each one is a standard Calc I integral.

i-component (power rule): $\int_1^4 2t^{3/2} dt = \frac{4}{5} t^{5/2} \Big|_1^4 = \frac{4}{5} (32 - 1) = \frac{124}{5}$

j-component (arctan): $\int_1^4 \frac{1}{1+t^2} dt = \arctan t \Big|_1^4 = \arctan 4 - \frac{\pi}{4}$

k-component (expand first, then power rule):

$$\int_1^4 (t+1)\sqrt{t} dt = \int_1^4 (t^{3/2} + t^{1/2}) dt = \frac{2}{5} t^{5/2} + \frac{2}{3} t^{3/2} \Big|_1^4 = \frac{256}{15}$$

$$\text{Result: } \frac{124}{5} \mathbf{i} + \left(\arctan 4 - \frac{\pi}{4} \right) \mathbf{j} + \frac{256}{15} \mathbf{k}$$

Your Turn: Evaluate $\int_0^{\pi/2} \langle \cos t, \sin t, 1 \rangle dt$

Integrate each component from 0 to $\pi/2$.

Try it before looking at the solution.

$$\int_0^{\pi/2} \langle \cos t, \sin t, 1 \rangle dt = \langle \sin t \Big|_0^{\pi/2}, -\cos t \Big|_0^{\pi/2}, t \Big|_0^{\pi/2} \rangle = \left\langle 1, 1, \frac{\pi}{2} \right\rangle$$

This vector represents the **net displacement** if a particle traveled with velocity $\langle \cos t, \sin t, 1 \rangle$ from $t = 0$ to $t = \pi/2$.

Homework

Section 13.2: 3, 5, 7, 9, 13, 19, 27, 29, 37, 39, 41